Sanjivani Rural Education Society's

Sanjivani College of Engineering, Kopargaon-423603

(An Autonomous Institute Affiliated to Savitribai Phule Pune University, Pune)
NAAC 'A' Grade Accredited

Department of Information Technology

UG Programme in IT NBA Accredited

FINAL YEAR PROJECT SYNOPSIS
(Academic Year 2026-2027)

Project Group Id	4
Title of the Project	An Intelligent NLP System for Legal and Employment Document Risk Analysis with Multiplatform Delivery
Project Domain	Artificial Intelligence, Natural Language Processing (NLP) , Large Language Models (LLMs) , Machine Learning ,Web Technologies
Problem Statement	Modern users frequently accept Terms of Service (ToS), Privacy Policies, Employment Agreements, and other legal documents without reading or understanding their contents due to their complexity and length. These documents often contain hidden clauses related to automatic renewal, data sharing, cancellation restrictions, third-party tracking, confidentiality, intellectual property, non-compete agreements, arbitration, hidden fees, and user liabilities. As a result, individuals unknowingly agree to terms that may expose them to legal, financial, or privacy risks. The proposed system aims to develop an AI-powered Legal and Employment Document Risk Analysis platform that automatically extracts text from uploaded documents or website URLs and analyses it using advanced Natural Language Processing (NLP) and Large Language Models (LLMs). The system identifies risky clauses, classifies them into Red Flags and Green Flags, computes an overall Trust Score and Risk Score, and generates simplified summaries in plain language. The application will provide recommendations to help users understand potential risks before accepting agreements. Additionally, the system supports multiple delivery platforms including a responsive web application and browser extension, enabling real-time document analysis during online sign-up or employment processes. This solution improves legal transparency, enhances digital awareness, and assists users in making informed decisions before signing legal or employment-related agreements.
References	[1] BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding, Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of NAACL-HLT*, 2019. [2] Language Models are Few-Shot Learners, Tom B. Brown *et al.*, "Language Models are Few-Shot Learners," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 1877–1901, 2020. [3] LEGAL-BERT: The Muppets Straight Out of Law School, Ilias Chalkidis, Manos Fergadiotis, Prodromos Malakasiotis, Nikolaos Aletras, and Ion Androutsopoulos, "LEGAL-BERT: The Muppets Straight Out of Law School," *Findings of ACL-IJCNLP*, 2020.

	[4] On the Opportunities and Risks of Foundation Models, Rishi Bommasani *et al.*, Stanford Center for Research on Foundation Models, 2024. [5] Attention Is All You Need, Ashish Vaswani *et al.*, "Attention Is All You Need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. [6] Lawformer: A Pre-trained Language Model for Chinese Legal Long Documents, Xiaoqi Cui *et al.*, "Lawformer: A Pre-trained Language Model for Legal Long Documents," *Artificial Intelligence and Law*, 2022. [7] CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review, John Hendrycks *et al.*, "CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review," *NeurIPS Datasets and Benchmarks Track*, 2021. [8] A Survey on Large Language Models, Wayne Xin Zhao *et al.*, "A Survey of Large Language Models," *ACM Computing Surveys*, 2023. [9] Constitutional AI: Harmlessness from AI Feedback, Anthropic, "Constitutional AI: Harmlessness from AI Feedback," 2022. [10] Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, Patrick Lewis *et al.*, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
Project Plan	*(Attach separate sheet specifying all phases of project)*
Name of Students in the Project Group with their signatures	1.Mr. More Sushant 2.Mr. Pawar Varun 3.Miss. Rahane Pratiksha 4.Miss.Shinde Payal
Name of the Guide and Signature	Mr. D. R. Cholke
Date	

Project title Accepted.

Mr.U.B.Sangule
Project Coordinator

Dr. M .A .Jawale
HOD IT